# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

# COMPUTER SCIENCE

Past Paper 2024 • Paper Code: PR–XII (A-I-A) (24)

Retyped faithfully from the original board paper — same questions, no changes.

## SECTION-A

**Time Allowed:** 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** PR–XII (A-I-A) (24)

ا۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. In____type of operating system does the processor remain idle when a process waits for an input.

(A) Multi Programming
(B) Batch Processing
(C) Real Time
(D) Time Sharing

Q2. When deleting a picture on your Mobile Phone, if the process stops for an Input Certificate ("Do you want to delete it?"), the process shifts from____to____.

(A) Running, blocked
(B) Blocked, Ready
(C) Running, Ready
(D) Blocked Running

Q3. A dispatchable unit of a process is called____.

(A) Task
(B) Program
(C) Thread
(D) Plug-in

Q4. A set of activities that are followed in an organized way for developing a software system is termed as____.

(A) Design
(B) SDLC
(C) Analysis
(D) Documentation

Q5. Pointer is a variable that stores____.

(A) Value
(B) Boolean Value
(C) Address of another variable
(D) IP address of computer

Q6. The ternary operator/conditional operator among the following is____.

(A) (? :)
(B) (; ;)
(C) (/ :)
(D) (@ :)

Q7. For single-line comments____is used.

(A) Single slash (/)
(B) Doubel slash (//)
(C) Slash and asterisk (/*....*/)
(D) Slash and asterisk (/*....*/)

Q8. In C++ language, for the Float Data type____bytes are reserved in memory.

(A) 8
(B) 4
(C) 2
(D) 01

Q9. For new line____escape sequence is used.

(A) /L
(B) /n
(C) /N
(D) /NL

Q10. Else body is executed when____.

(A) Else condition is true
(B) If condition is true
(C) If condition is false
(D) None of these

Q11. The default statement in switch statement is____.

(A) Mandatory
(B) Optional
(C) Enforced
(D) Cannot

Q12. Statement terminator in C++ is____.

(A) Colon (:)
(B) Semi colon (;)
(C) Comma (,)
(D) Full stop (.)

Q13. The____header file shall be included for file handling.

(A) iostream
(B) conio
(C) fstream
(D) filestream

Q14. The operator used for finding the size of an array is____.

(A) Length ()
(B) Size of ()
(C) Array size ()
(D) Count of ()

Q15. Concatenating or joining two strings together____built-in function can be used.

(A) string cat ()
(B) strcat ()
(C) string cocat ()
(D) str concat ()

## SECTION-B

**Time Allowed:** 2 Hours 40 Minutes | **Maximum Marks:** 40

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q2. Answer any TEN parts. Each part carries FOUR marks.

1) Define operating. List down at-least three functions of operating system.

2) Elaborate Multiprocessor Operating System. Explain its advantages.

3) Explain the 5-states process model that an operating system applies to manage the execution of processes.

4) Differentiate between Multitasking and Multiprogramming Operating System.

5) Consider the following algorithm. Draw flowchart representing the steps outlined in the algorithm. (i) Read three (03) inputs from user (ii) Add the values (iii) Average = sum/3 (iv) Display the Average

6) Write down at-least four differences between the constructor and destructor.

7) Describe the setw manipulator in C++. Specify the header file required for setw and demonstrate its usuage with C++ program.

8) Discuss Member Access Specifiers in C++ language.

9) Define Function in C++ list three essential components for using the user-defined function.

10) Describe the concept of inheritance in C++ language and provide a C++ program to illustrate its implementation.

11) Discuss if-else structure. Write its syntax; create a flowchart for if-else.

12) Write a C++ program with a user-defined function named sum(), for adding two values and returns the result to main () functions.

13) Discuss function overloading. Provide an example in C++ language.

## SECTION-C

**Time Allowed:** 2 Hours 40 Minutes | **Maximum Marks:** 20

Note: Note: Attempt any TWO questions. All questions carries equal marks.

Q. Attempt any TWO questions. All questions carries equal marks.

Q3.

(a) Define loop in C++ language. Write syntax for while loop and for loop.

(b) Write a C++ program to display the multiplication table of a given number upto ten (10). Sample: 2×1 = 2 2×2 = 4 – – – – – – 2×10 = 20

Q4.

(a) Define Array and discuss its two main types. Describe how to declare and initialize both types.

(b) Write a C++ program based on the following instructions. (i) Take Six (06) integer values from user (ii) Store them in Array (iii) Display the sum

Q5.

(a) Define class and object.

(b) Write a C++ program with the following specification; (i) Create a class "Marks" with three data members for storing marks of three subjects. (ii) Create three member functions; (a) Input ( ) to enter marks of 3 subjects (b) Sum ( ) to addd and return total marks (c) avg ( ) to find and return average of three subject marks